3 **(a)** **(i)**

$$v_{max} = 1.4 \text{ m s}^{-1}$$

$$E_K = \frac{1}{2} m v_{max}^2$$

$$= \frac{1}{2}(0.50)(1.4)^2 \qquad \text{[C1]}$$

$$= 0.49 \text{ J} \qquad \text{[A1]}$$

(ii)

$$v_{max} = \omega x_0$$

$$x_0 = \frac{1.4}{\left(\frac{2\pi}{0.50}\right)} \qquad \text{[C1]}$$

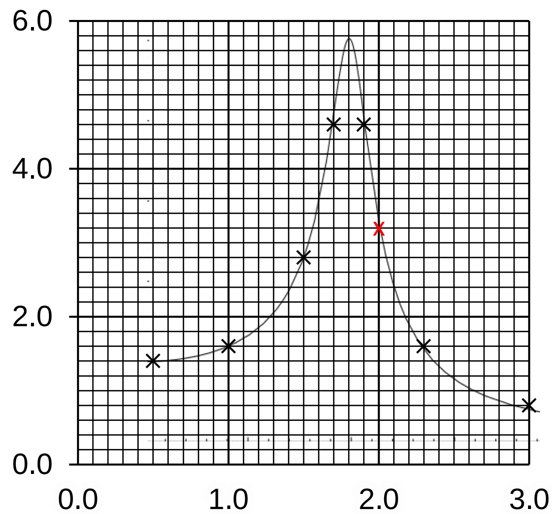
$$= 0.11 \text{ m} \qquad \text{[A1]}$$

(b) Damped oscillation. [B1]

Air resistance [B1] on the wooden board.

(c) **(i)**

(ii)



Correct plot (2.0, 3.2) [B1]

Correct curve of best fit [B1]

(iii) 1.8 Hz (accept 1.75, 1.8, 1.85) [B1]

4 (a) (i)